

ASSIGNMENT 4

WATERMARK FORGERY ATTACK

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1 INTRODUCTION

The digital watermarking technology serves to protect copyrights, prove ownership and establish the authenticity of digital objects using embedded information which cannot be detected by naked eye. Unfortunately, this technology may face certain threats related to the forging of watermarks. A potential forger uses the technique of watermark forging attack by transferring watermarks from one set of images into another, making forged images seem like legitimate ones. Our aim in this assignment was to carry out a watermark forging attack using black-box model, which means that we did not have any information about the used watermarking scheme. What was required was to take a set of watermarked source images and a clean target image set and make modifications to the targets in order to embed the watermark into them while maintaining their high quality.

2 APPROACH

Our approach consists of two stages: watermark extraction and watermark injection.

2.1 WATERMARK EXTRACTION & INJECTION

For each group of watermarks, the estimated value of the embedded watermark was extracted from a set of watermarked images given.

Instead of estimating the watermark directly, we first used the artifact discriminator for watermarks based on WMForger, i.e., the preference model, which was pre-trained on ConvNeXt architecture. This optimization procedure is aimed at removing artifacts due to watermarks and preserving the content of the image. We optimized the perturbation in the learnable manner using gradient ascent for 300 optimization iterations with the learning rate of 0.05. For each clean target image, the extracted watermark residual was resized (if necessary) to match the target image dimensions. The extracted residual was scaled using the $\alpha = 1.5$ before being added to the target image. Finally, pixel values were clipped to the valid image range [0,255], converted to unsigned 8-bit integers, and saved with their original filenames. The resulting 200 forged images were packaged into a ZIP file for submission.

2.2 RESIDUAL FILTERING

However, even after extraction of the watermarked residual there were still traces of low-level noise. In order to reduce noise we kept only those watermark residuals with high strength. We took the absolute value of all watermark residuals and only kept the top 30%, that is we took out everything below the 70th percentile threshold and set them to zero.

2.3 EXPERIMENTAL EVALUATION

We evaluated two watermark forgery approaches during the development of our solution. All experiments were evaluated on the public leaderboard, with the leaderboard score used as the primary performance metric.

k	n	α	Allow	Score
50	5	1.0	–	0.279
50	5	0.9	–	0.347
150	10	1.5	–	0.459
150	10	2.0	–	0.462
300	10	1.5	–	0.462
300	10	1.5	70	0.479
500	10	1.5	70	0.479

Table 1: Hyperparameter configurations. Here, k is no. of optimization steps, n is no. of source images used for watermark extraction, and *Allow* is threshold for residual sparsification.

Averaging-based Residual Attack. As our starting point, we carried out averaging attacks using an estimate of the watermark residual as the difference between the averages of watermarked images and clean images. The first method of our implementation using pixel mixing yielded a score of about 0.03. Performance improved considerably after correction of the watermark residual estimate, and after adjusting the strength of watermarking separately for each technique, a final score of **0.214** was obtained. Though it was cheap to compute, the performance was limited by the inability to separate the watermark from the rest of the image using averaging techniques.

Preference-Model-Based Attack. We then used the preference model to implement the watermark forger using the authors’ freely available codebase along with the pretrained artifact discriminator based on ConvNeXt-Tiny. The extraction was done using gradient ascent optimization technique and then transferring of the extracted residual to the target images. Instead of changing steps (k) we added a scalar multiplier α to scale the residual. We observed the quality and detection strength tradeoff.

3 RESULTS

This method was successfully able to forge images out of all 200 target images without any loss of visual appearance.

The use of the pre-trained artifact discriminator helped to extract watermarks without having any knowledge about the watermarking algorithm used. The use of median aggregation made this extraction more robust as the image-dependent artifacts could be suppressed. Moreover, the percentile filter was applied to eliminate weak noise signals, hence making it possible to inject stronger watermark signals into the clean images. An increase in the watermark strength factor from $\alpha = 1.5$ led to improvement in the watermark detection score but reduced the visual quality.

4 CONCLUSION

In conclusion, the current experiment shows that an invisible watermarking system may prove to be vulnerable to forgery attacks without prior knowledge of the watermarking algorithm used. Watermark residual extraction from several watermarked pictures followed by transferring it to clean pictures makes it possible to forge pictures with a retained high visual quality that contain detectable watermark data. The problem discussed above proves to have certain consequences for copyright protection, authentication of digital media. If it turns out to be possible to forge the marks of ownership, the validity of the watermark-based verification systems will become questionable.

5 REFERENCES

1. Can Simple Averaging Defeat Modern Watermarks?
2. Transferable Black-Box One-Shot Forging of Watermarks via Image Preference Models
3. Github Link
4. Preference Model Github